

Addanki Venkata Navanth Sai Teja

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EDUCATION

Indian Institute of Information Technology, Allahabad
B.Tech in Information Technology — CGPA: 8.05

Aug 2023 – May 2027
Prayagraj, Uttar Pradesh

EXPERIENCE

Research Intern

Indian Institute of Information Technology, Allahabad

May 2026 – July 2026

Prayagraj, India

- Researched **gloss-free Indian Sign Language Translation** using a **Dual-Head Transformer** architecture for semantic and phonological prediction.
- Developed a sign-language processing pipeline using **MediaPipe Holistic** to extract 3D skeletal landmarks capturing hand gestures, body posture, and facial expressions.
- Implemented a **1D-CNN** for temporal compression and a shared Transformer encoder with parallel **English and IPA prediction heads**, trained using a weighted multi-task loss.
- Worked with the **iSign dataset** and evaluated translation performance using **BLEU-4** and **ROUGE**, achieving a **23.66 BLEU-4 score**.

PERSONAL PROJECTS

ApexFit – AI-Powered Fitness SaaS | *React, TypeScript, Node.js, PostgreSQL, Redis, pgvector, Ollama, Cohere*  2026

- Built a full-stack fitness SaaS with AI coach chat, workout planner, macro tracker, and Stripe subscription billing using **React + Vite** (frontend) and **Node.js + Express + PostgreSQL + Redis** (backend) with JWT auth.
- Implemented a **Daily Briefing** system that surfaces the least-recently-trained muscle group using real workout history, computes protein remaining against a Mifflin-St Jeor TDEE-based target, and grounds dinner suggestions in live **USDA FoodData Central** API data.
- Built a two-stage **RAG pipeline** using **pgvector HNSW** (ANN retrieval) and **Cohere rerank-english-v3.0** cross-encoder re-ranking (top-4 selection), with per-user profile snapshots re-embedded on every meaningful data change for retrieval freshness.
- Designed a **deterministic plateau root-cause engine** correlating training volume, protein intake, and recovery against the user's own baseline using adaptive time windows; and a **6-stage workout recommendation pipeline** with injury-safe filtering, regression-based ranking, and double-progression overload — LLM used solely for explanation.
- Applied **Redis caching** across 5 endpoints and an atomic **INCR-based rate limiter** for free-tier AI coach quota; all paths fail open gracefully.

CampusOS | *Next.js, FastAPI, PostgreSQL, Neonbase, Gemini AI, Groq Vision*



- Developed a full-stack **AI-powered Campus Operating System** for **Amazon HackOn 6.0** using **Next.js, FastAPI, PostgreSQL, Neonbase, and Gemini AI**.
- Built an intelligent campus assistant with **role-based access control (Student, Faculty, Admin)**, enabling natural language queries, attendance prediction, notices, placements, and timetable management.
- Implemented **OCR-based timetable and mess menu extraction**, AI-powered scheduling, attendance analytics, and conflict detection using **Groq Vision** and **Gemini**.
- Designed secure **JWT authentication**, REST APIs, and deployed the application on **Render**, providing a scalable and responsive campus management platform.

TECHNICAL SKILLS

Languages: C, C++, Java, JavaScript, SQL, HTML, CSS

Frameworks: ReactJS, Node.js, Express.js, Next.js, FastAPI

AI/ML: RAG, LLM Inference, Ollama, pgvector, Cohere, Gemini AI, Groq Vision

Databases: PostgreSQL, MySQL, MongoDB, Redis, Prisma ORM

Core CS: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, OOP

Developer Tools: Git, GitHub, VS Code, Cursor, DevOps, CI/CD

ACHIEVEMENTS

- Selected for the **Reliance Undergraduate Scholarship 2023** for academic excellence.
- Selected among the **top 0.5%** (Stage 2) of **20,000+** applicants in the **AlgoUniversity Technical Fellowship 2024**.
- Achieved **Specialist** rating on **Codeforces**, **3-Star** rating on **CodeChef**, and **Knight** rating on **LeetCode**, with **500+ problems solved** across competitive programming platforms.
- Secured **Rank 1420** in **IICPC CodeFest'25 Prelims** among **45,000+** participants.
- Selected as a **Semi-Finalist** in **Flipkart GRiD 8.0**.
- Ranked in the **top 0.9%** in **JEE Mains 2023**.